- Separated OMat24 into AIMD and Rattled subsets for training (different data ratios and loss coefficients)
- Additional data
  - Updated OMol25 + OPoly26[44]
  - OC22 [72]
  - OC25 [61]
  - Diatomic data for OMol25 and OMat25
  - Single atom data for all datasets

We will provide further details of UMA-1.2 changes in the near future. All the updated code and configuration files will be available at release in the fairchem repository.

Table 9 Summary of main training-related hyper-parameters for UMA-S-1.2

Hyper-parameter	Stage 1	Stage 2
Precision	FP32	FP32
Radius cutoff Å	6	6
Max neighbors	300	300
Force prediction	Autograd	Autograd
Stress prediction	Autograd	Autograd
Optimizer	AdamW	AdamW
Learning rate scheduling	Cosine	Cosine
Maximum learning rate	8×10^{-4}	4×10^{-4}
Warmup epochs	0.01	0.01
Warmup factor	0.2	0.2
Gradient clipping norm threshold	100	100
Model EMA decay	0.999	0.999
Weight decay	1×10^{-3}	1×10^{-3}
Number of MoLE experts	64	64
Number of layer blocks	4	4
Maximum degree L_{max}	2	2
Maximum order M_{max}	2	2
Number of channels N_{channel}	128	128
Number of radial basis functions	32	32
Global batch size (atoms)	88k	88k
Number of epochs	2	1

Table 10 Overview of the datasets used to train UMA-S-1.2.

Dataset	Sampling ratio
OMat24 (AIMD)	4
OMat24 (Rattled)	2
OMol25+OPoly26	16
OC20++	2
OMC25	4
ODAC25	2
OC22	2
OC25	16

B Training Data

A summary of the five datasets used to train UMA is shown in Table 11. In total, the dataset has 459 million training examples, containing up to 350 atoms. The average number of atoms varies based on the dataset, from 19 for OMat24 to 178 for ODAC25.

Table 11 Overview of the five datasets used to train UMA. For each dataset various statitiscs are provided alongside the sampling ratio used for training.

Dataset	Domain	Training Size	Labels	# Elements	Avg Size	Force RMS	Sampling ratio
OMat24	Materials	100,824,585	E,F,S	89	19	2.83	4
OMol25	Molecules	75,889,983	E,F	83	52	0.985	4
OC20++	Catalysis	229,054,043	E,F	56	77	0.624	1
OMC25	Molecular Crystals	24,870,226	E,F,S	12	130	0.103	2
ODAC25	MOFs	28,517,826	E,F	70	178	0.046	1
Total		459,156,663					

B.1 Materials

The field of inorganic bulk materials is moving at an incredibly fast pace. Here we train on the Open Materials (OMat24) dataset (100M) [5], one of the largest and most diverse datasets in the community. All DFT calculations from this domain were run with VASP [39–42] and used the PBE [55] functional. Due to the differences in psuedopotential version and different pseudopotentials for certain elements in OMat24 and Materials Project [5] calculation settings used for the data in most third party benchmarks, finetuning was also performed on MPtrj [17] and subsampled Alexandria (sAlex) [62] to ensure consistent evaluation on the materials benchmarks.

B.2 Molecules

The community has seen dozens of molecular datasets spanning different scales for a variety of applications []. However, the varying levels of DFT theory and quality makes it challenging to unify under a single model. The release of the Open Molecules 2025 (OMol25) dataset [45] helps address this by providing the largest single dataset (100M+) spanning 80+ elements covering metal-complexes, biomolecules, electrolytes, and several existing datasets under a single, high-quality level of theory. All DFT calculations were performed using Orca [51] (ωB97M-V/def2-TZVPD). At the time of training, only 75M samples from OMol25 were available for use, and we refer to this as OMol-preview. Splits were constructed to ensure that this snapshot of the dataset is consistent with the full dataset release. All ablations and results were trained with this OMol-preview, unless stated otherwise. Released models, however, will be retrained with the full OMol25 dataset to ensure the best models are accessible by the community.

B.3 Catalysis

The Open Catalyst (OC20) dataset [11] provides the largest adsorbate+surface dataset in the community. OC20 enumerates 1M+ unique surface + adsorbate combinations, spanning 55 elements, and runs local geometry optimizations. Here, we train on the OC20 All (133M), MD (38M), and Rattled (17M) datasets. Unlike prior work, we also leverage OC20's clean surface data (14M) since models here are trained on total energies. One limitation of OC20 is that it only contains single adsorbates that interact with a surface. To address this, we introduce the OC20-Multi-Adsorbate (mAds) dataset (22M) to better capture coverage effects and adsorbate-adsorbate interactions. All DFT calculations were performed using VASP [39–42] with the RPBE exchange-correlation functional functional [27].

B.4 Molecular Crystals

The most recent 7th Crystal Structure Prediction (CSP) Blind Test organized by Cambridge Crystallographic Data Center (CCDC) demonstrated the effectiveness of tailored machine learning interatomic potentials

(MLIPs) in predicting, filtering, and ranking molecular crystal structures [30, 31]. However, despite the widespread applications of molecular crystals, there has been limited focus on developing universal MLIPs for molecular crystals, mostly because of the scarcity of publicly available datasets. Currently, publicly available datasets of molecular crystals are scarce, with at most 60,000 materials represented [3, 9]. To address this data gap, we use the Open Molecular Crystals (OMC25) dataset, which comprises 25 million molecular crystal structures. The dataset includes multiple relaxation trajectories of various molecular crystal packings generated with Genarris [71] starting from organic molecules from the OE62 dataset [69]. The dataset includes 12 elements (all elements from OE62, excluding Li, As, Se, and Te) and the maximum number of atoms is capped at 300. The dataset is computed using VASP [39–42] with the PBE exchange-correlation functional [55] and D3 dispersion correction [26]. The OMC25 dataset, along with in-depth details on data and polymorph structure generation, will be released in an upcoming publication [19].

B.5 Metal-organic frameworks (MOFs)

The Open Direct Air Capture 2025 (ODAC25) dataset represents the largest MOF dataset (78M) for DAC applications [20]. ODAC25 extends the earlier ODAC23 [68] dataset, and is derived from the CoREMOF [13, 14] dataset. ODAC25 focuses on the adsorption and co-adsorption of CO_2 and H_2O in MOFs among other adsorbates, which is critical for DAC performance evaluation. This work uses the subset of ODAC25 that overlaps with ODAC23, where all DFT-computed adsorption energies have been upgraded to a higher k-points sampling grid density, providing improved accuracy over ODAC23 (29M calculations). All DFT calculations were performed using VASP [39–42] with the PBE exchange-correlation functional [55] and D3 dispersion correction [26].

C Scaling Laws Methods

The scaling law experiments were only performed on pretraining; due to compute constraints, we did not study the effect on finetuning with energy conservation. We used an 8-expert MoLE version of the model with the UMA-M settings ($l_{max} = 4$, $m_{max} = 2$, and $n_{neighbors} = 30$) for consistency.

C.1 FLOP counting

We approximate our training FLOPs by

$$C(N, D) \approx \kappa N D, \tag{3}$$

where N is the total number of parameters in the network, and D is the number of inputs (tokens for LLMs and atoms or edges for MLIPs) computed on. This relationship holds for any network that is dominated by linear layers where κ indicates how many times a parameters is re-used on an input. For a single forward pass of a linear network, $\kappa = 2$. A full training cycle for such a network with a single backward pass brings $\kappa = 6$ as we need to compute the gradient with respect to both the parameters and inputs. Thus, for most LLMs $\kappa = 6$ FLOPs/parameter/token [36] for a single forward and backwards pass where D is in units of tokens.

For our edge-based SO2 equivariant networks [54], κ is a function of l_{max} and m_{max} spherical harmonic orders and the number of edges per atom $n_{neighbors}$. For the scaling law experiments, we used $l_{max} = 4$, $m_{max} = 2$, and $n_{neighbors} = 30$ which corresponds to the settings of UMA-M model, resulting in $\kappa \approx 270$ FLOPs/parameter/atom (or 9 FLOPs/parameter/edge) per training step, which can be computed or experimentally determined. As long as the number of input edges and parameters are sufficiently large (which holds for UMA models), this flop approximation holds as contributions from all other operators are marginal. We also verified this assumption with direct FLOP counting in our model code.

Overall, parameter reuse is significantly higher in Equivariant-GNNs compared to LLMs, and hence the flop count is 2-3 orders of magnitude higher for a similar parameter-sized LLM network.

C.2 Compute optimal fits

The compute optimal model and dataset sizes can then be fitted to power laws [29, 36]:

$$\begin{aligned}\log(N^*(C)) &= \alpha \log(C) + A \\ \log(D^*(C)) &= \beta \log(C) + B\end{aligned} \tag{4}$$

Where C is the compute in FLOPs described in Appendix C.1. $N^*(C)$ represents the optimal model size (in parameters) as a function of compute, and $D^*(C)$ represents the optimal dataset size (in units of atoms). $N^*(C)$ is determined by finding the minima of fitted parabolas for each Isoflop curve. The 10% and 90% percentile bootstrap errors are shown in Figure 3(e). α and β are the scaling coefficients w.r.t. model size and dataset size respectively. A and B are offset constants of the fit. Fit coefficients and bootstrap errors are shown in Table 12.

C.3 Fitting loss vs. parameters

To understand the minimum achievable loss for dense vs MoLE models we can fit the more general parameterized ansatz of $L(N, D)$ proposed by [36].

$$\tilde{L}(N, D) = \hat{E} + \frac{\hat{A}}{N^{\hat{\alpha}}} + \frac{\hat{B}}{D^{\hat{\beta}}} \tag{5}$$

This maps the power law coefficients from Equation 5 to those in Equation 4 by using $\alpha = \frac{\hat{\beta}}{\hat{\alpha}+\hat{\beta}}$ and $\beta = \frac{\hat{\alpha}}{\hat{\alpha}+\hat{\beta}}$. Here we can either minimize the $\tilde{L}(N, D)$ directly by fitting the 5 parameters $\hat{E}, \hat{A}, \hat{B}, \hat{\alpha}, \hat{\beta}$ with a iterative minimization procedure such as LBFGS [10, 29] or by examining the loss as a power relationship of N^* using

$$\log \tilde{L}(N^*) = \hat{\alpha} \log(N^*) + \gamma \tag{6}$$

where $\gamma = log([1 + \frac{\hat{\alpha}}{\hat{\beta}}]\hat{A})$ with $\hat{E} \approx 0$. We found both methods yielded similar results but the minimization of $\tilde{L}$ was more sensitive to the choice of hyperparameters.

Table 12 Power Law Coefficients determined from fitting Equations 4 and 6. Error bounds are determined by bootstrap sampling 1000 times and taking the 10th and 90th percentile values, quoted in brackets.

Parameter	Dense	MoLE
α	0.61 (0.57,0.65)	0.56 (0.49,0.59)
β	0.39 (0.35, 0.43)	0.44 (0.39, 0.43)
A	-4.5 (-3.8, -5.3)	-3.8 (-2.56, -4.65)
B	3.6 (2.9, 4.4)	2.9 (1.6, 3.7)
$\hat{\alpha}$	-0.29 (-0.27, -0.31)	-0.25 (-0.2, -0.3)
γ	2.16 (2.02, 2.34)	1.82 (1.61, 2.12)

D Inference

For inference benchmarking we use a periodic fcc carbon system with lattice constant $a = 3.8$Å. This results in a fixed density of approximately 50 edges per atom within 6Å. For UMA models, we use a combination of torch.compile, cuda graphs and pre-merged MoLE experts for inference speed. For large number of atoms (> 1000), we use edge-based activation checkpointing to trade off memory for some speed, allowing us to fit 100k+ atoms for the UMA-S into memory. We checked all our optimizations chosen does not degrade simulation accuracy, equivariance or energy conservation properties. While our benchmarks do not include graph generation, our internal CUDA based graph generation algorithm is very fast and decreases throughput by no more than 10% even for the largest systems tested. In the case of non-MoLE merging, we found the inference speed was comparable but the parameters require more GPU memory to store.

For fair comparisons against other models, we used pytorch2.6.0, cuda12.4, python3.12 and TF-32 precision universally on a H100 80GB GPU. We use standard torch.compile settings whenever possible (only MACE-MPA-0 failed to compile). Different models have different radius cutoffs and max neighbors settings. We made sure that all models was receiving roughly 50 neighbors per atom for the same number of atoms.

Update March 2026 - UMA-S-1.2 requires fairchem-2.16.0 to run.

For test systems with 100 and 1000 atoms, we introduced a new inference optimized setting specifically for UMA-S architecture called `execution_backend=umas_fast_gpu`, this is now the default setting in `turbo` mode.

E Evaluations

Table 13 Test MAE results on held out test splits for materials [59], catalysis [11], molecules [45], molecular crystals [19] and ODAC [68]. All energies are in meV, forces are in meV/Å and stresses are in meV/Å^3. Results for UMA are compared against the SOTA literature results when available and other strong baselines trained only on the domain-specific dataset. Target accuracies for practical utility are provided as an approximate guide for reference.

	Materials						*Catalysis*				*Molecules*				*Molecular crystals*			*ODAC*	
Model	WBM *Energy/Atom*	*Forces*	*Stress*	HEA *Energy/Atom*	*Forces*	*Stress*	ID *Ads. Energy*	*Forces*	OOD-Both *Ads. Energy*	*Forces*	OOD-Comp *Energy/Atom*	*Forces*	PDB-TM *Energy/Atom*	*Forces*	OMC-Test *Energy/Atom*	*Forces*	*Stress*	OOD-L/T *Ads. Energy*	*Forces*
UMA																			
UMA-S	20.0	60.8	4.4	22.0	72.8	3.1	52.1	24.3	70.2	30.9	3.64	10.80	0.88	16.12	0.91	4.77	0.97	292.4	16.0
UMA-S-1	19.4	62.2	4.5	21.9	73.5	3.1	53.1	24.5	70.4	31.2	0.96	8.25	0.93	15.56	0.93	5.15	1.01	287.1	13.6
UMA-S-1.1	20.2	62.8	4.4	24.9	83.7	3.5	51.5	24.1	68.8	30.7	0.95	8.64	0.84	15.47	1.03	5.04	0.93	289.9	13.3
UMA-M-1.1	18.2	50.7	4.2	21.9	69.0	3.5	31.8	15.5	45.5	20.2	0.74	5.44	0.50	10.14	0.84	2.83	0.90	294.1	10.3
Literature																			
eSEN-OMat [23]	16.2	49.6	4.1	20.0	59.5	3.2	-	-	-	-	-	-	-	-	-	-	-	-	-
eqV2-OMat [5]	14.9	46.3	3.6	20.3	47.0	2.7	-	-	-	-	-	-	-	-	-	-	-	-	-
eqV2-OC20 [46]	-	-	-	-	-	-	149.1	11.6	306.5	15.7	-	-	-	-	-	-	-	-	-
GemNet-OC20 [24]	-	-	-	-	-	-	163.5	16.3	343.3	23.1	-	-	-	-	-	-	-	-	-
eSEN-sm-cons. [45]	-	-	-	-	-	-	-	-	-	-	1.35	7.39	0.83	12.72	-	-	-	-	-
eqv2-ODAC [68]	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	316.0	7.2
ST Baselines																			
eSEN-S-OMol	-	-	-	-	-	-	-	-	-	-	3.67	11.56	0.79	14.11	-	-	-	-	-
eSEN-S-OMC	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.05	5.39	0.94	-	-
Target																			
Practical Utility	10-20	-	-	10-20	-	-	100	-	100	-	1-3	-	1-3	-	1-3	-	-	100	-

E.1 UMA-1 and 1.1 Results

In this section, we provide results on the UMA-1 and UMA-1.1 models trained with the entire OMol25 dataset as released (05/14/2025), instead of the preview OMol25 dataset used by UMA in the main text of this paper. The other datasets used for training were unchanged, with the exception of the sampling ratios (Table 11) where the OMat24 coefficent was changed from 4 to 2 for both UMA-1 and UMA-1.1. We provide UMA-S-1.1 and UMA-M-1.1 results, which resolved a size extensive bug later discovered. These models were fine-tuned on our 1.0 models with the fix in place. Tables 13 and 14 correspond to Tables 2 and 4 in the main text.

E.2 Materials

Full results on materials' benchmarks are in Tables 15 and 16. Table 15 shows both validation and test results following OMat24 [5] along with the new high entropy alloy HEA test introduced in this paper. The HEA dataset contains relaxation trajectories for over 5000 alloys with atomic configuration disorder. Input structures were generated by sampling metallic element combinations of up to 6 different unique elements and using the special quasirandom structure method [4, 79] to decorate face-centered cubic, body-centered cubic and hexagonal close packed structures. DFT relaxations were carried out following the settings used in the OMat24 dataset [5].